KIDWAI MEMORIAL INSTITUTE OF ONCOLOGY
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DEPARTMENT OF QUALITY AND PATIENT SAFETY

NQAS & NABH EL 2nd EDITION KEY PERFORMANCE AND OUTCOME INDICATORS AS APPLICABLE TO KMIO

KEY PERFORMANCE INDICATORS								
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Bed Occupancy Rate	Total Patient bed days (Midnight head count of each day added for the month of all patients) Day Care Patients	Product of Total number of functional beds in the hospital and days in the month Exclusion:- Observation Beds	(Total Patient bed days *100/Functional beds*days in month)	Monthly	Mid night census	Indicator for Utilization of Hospital Indoor services
	2	Lab test done per thousand patients	Total number of tests done for both OPD and IPD patients Exclusion - Test done at Point of care	Total number of patients attended during the month Inclusion:- Both OPD and IPD cases	(Total number of lab tests done*1000/Total number of patients attended)	Monthly	Lab Register	Indicator to measure Utilization of laboratory services

	3	Percentage of surgeries done in night out of total surgeries	Total major surgeries conducted during night (8 PM to 8 AM) Exclusion – Minor Surgeries	Total major surgeries conducted in Hospital (Day+Night) Exclusion – Minor Surgeries	Total number of major surgeries conducted in night time*100/Total number of major surgeries conducted	Monthly	OT Register	Preparation of OT at night time for major surgeries
	4	Percentage of surgeries done during day out of total surgeries	Total number of planned major surgeries conducted during day time (8 AM TO 8 PM)	Total major surgeries conducted in Hospital (Day+Night)	Total number of planned major surgeries conducted*100/Total number of major surgeries	Monthly	OT Register	Preparation of OT for planned surgeries
Efficiency	5	Emergency Death Rate	Total number of deaths in emergency Exclusion:- Brought dead	Total number of registered patients in emergency Exclusion - Cases referred out	Total number of deaths in emergency*100/Total number of registered patients in emergency	Monthly	Emergency register	To check efficiency of Emergency department
	6	Major Surgeries per surgeon	Total number of major surgeries conducted	Total number of surgeons appointed in the facility	Total number of major surgeries conducted/Total number of surgeons appointed	Monthly	OT Register	Indicator for efficiency of Surgeons
	7	OPD Per doctor	Total number of patients attended in OPD	Total number of doctors available in the hospital Inclusion: Regular, contractual, Part Time Exclusion:- Doctors not engaged in OPD	Total number of Patient consulted in OPD/Total number of doctor appointed for OPD	Monthly	OPD Register	Indicator for measuring efficiency of Doctors in OPD

				like MS, Radiologist, Microbiologist				
	8	External Quality Score for Lab tests(Median Value)			Take Median of all scores obtained (Arrange scores in increasing order- Pick the middle value if numbers are odd- Take average of middle two values if numbers are even)	Monthly	Lab EQAS register	Indicator to measure efficiency of Lab
	9	Average length of stay	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Total number of discharges. Inclusion:- Normal discharge, LAMA, Abscond, Referral, deaths	Total Patient bed days /Total Discharges	Monthly	Admission and Discharge Register	Indicator of Quality of Clinical care and infection control practices
	10	Surgical site infection rate	Total number of Surgical site infection detected (Any purulent discharge, abscess, spreading cellulites at surgical site during the month after the surgery)	Total number of surgeries conducted (major & minor surgeries)	(Total number of surgical site infection detected*100/Total number of surgeries Conducted)	Monthly	Infection control monitoring report	Indicator for Quality of infection control practices in OT
	11	Blood Replacement Rate	Total no. of Blood Unit issued on replacement in each day added for Month	Total no of blood unit issued in the month Inclusion- Blood Unit issued without replacement	Total number of blood unit issued on replacement*100/Total number of blood units issued	Monthly	Blood issue register	Indicator of availability of blood from voluntary donation

			Exclusion – Blood Units issued without replacement					
Service Quality Indicator	12	Left against Medical advice (LAMA) Rate	Total number of LAMA patients from the facility Exclusion:- Abscond and referral cases	Total admission in the facility	(No. of LAMA Patients from the facility*100/Total no. of admission)	Monthly	Admission /Discharge Register	Indicator of service quality and patient satisfaction with treatment and stay in IPD
	13	Patient Satisfaction Score (IPD)	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each patients in Patient satisfaction survey for indoor patients done each month on statistically adequate sample (at least 30)	Monthly	Record of Patient Feedbacks	Indicator of patient satisfaction in IPD
	14	Patient Satisfaction Score (OPD)	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each patients in Patient satisfaction survey for outdoor patients done each month on statistically adequate sample (at least 30)	Monthly	Record of Patient Feedbacks	Indicator of patient satisfaction in OPD

	15	Registration to Drug time			Average time taken by a patient from entering in queue for OPD registration to finally getting drugs at Pharmacy counter observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30)	Monthly	Time Motion Study	To observe waiting time for OPD procedures
COP.3.b	16	Mortality Rate	Number of Deaths	Number of Discharges and deaths	(Number of Deaths/Number of Discharges and deaths)*100	Monthly		

OUTCOME INDICATORS							
	1.Emergency Department						
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	No. of Emergency cases per thousand population	Total number of emergency cases treated or managed in A& E department.	Total population in defined geographical area of the DH	(Total no. of cases attended by A&E department of the hospital *1000/Total Population)	Monthly	A& E Register	Utilization of A&E department of Hospital
	2	No. of trips per ambulance	Trips made by hospital owned ambulance for transportation of patients Exclusion:- ambulance trips used for other purpose like transporting of on call doctors, carrying medicines, camp duty, Drop back under JSSK	Exclude 108 or other ambulance	Number of trips done by ambulance/ Total Number of ambulance available in the facility	Monthly	Log Book of Ambulance	Utilization of Hospital ambulances
	3	No. of resuscitation done per thousand population	Total number of cases where resuscitation done Inclusion:- chest compression, airway and breathing	Total population in defined geographical area of the DH	(Total number of emergency cases managed by resuscitation*1000/Total Population)	Monthly	Case sheet	Indicator to see preparedness for emergency response

	4	Proportion of patients attended in night	Total number of patients attended in A&E department from 8PM to 8 AM	Total Number of patients attended in A&E department	(Total Number of Patient attended during night time/Total Number of patient attended in A& E Department)	Monthly	A& E Register	Indicator to see preparedness & Utilization of A&E department at night
	5	Proportion of BPL patients	Total number of BPL Patients Inclusion:- Patients having BPL card or equivalent card	Total Number of patients attended in A&E department	(Total Number of BPL Patient attended in A& E department/Total Number of patient attended in A& E Department)	Monthly	A& E Register	Indicator to see Utilization of A&E department by BPL patients.
Efficiency	6	Response time for ambulance	Response time= Response time elapsed between receiving of call and actual arrival of ambulance.		Mean of all response time taken added for all ambulances	Monthly	Log Book of Ambulance	Indicator for measuring efficiency of ambulance services
	7	Proportion of cases referred	Total number of cases which referred from A&E department to higher facility. Exclude:- LAMA, Abscond patients	Total Number of patients attended in A&E department	(Total number of cases referred out from A&E department/Total number of cases attended in A&E department)	Monthly	A& E Register/Referral Register	Indicator for measuring efficiency of clinical services

	8	Response time at emergency for initial assessment	Response time=Time taken from the moment the patient entered in A&E department to the moment the patient's assessment done by emergency medical officer		Mean of response time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30)	Monthly	Time motion study	Indicator for measuring efficiency of emergency response
	9	Average Turn around time	Time elapsed between moment patient entered in A&E department to the shifting of patient from A&E department		Mean of turn around time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30)	Monthly	Admission Register and transfer register of A& E Department	Indicator for measuring efficiency of the department
	10	Proportion of patient referred by state owned/108 ambulance per 1000 referral cases	Total number of cases referred through State owned/Facility owned/108 ambulance	Total number of cases referred from A&E department	(Total number of patients referred by state owned or 108 ambulance*1000/Total number of referral cases from A&E department)	Monthly	Referral register of A&E department	Indicator for measuring efficiency of hospital referral services
Clinical care and safety	11	No of adverse events per thousand patients	Total number of adverse events Inclusion:- Patients fall, Medication error, Needle stick injury	Total number of patients attended in A&E department	(Total number of incidences of adverse events*1000/ Total Number of patients attended in A& E)	Monthly	Adverse event form/Register & Admission register	Indicator to measure safe clinical practises in A&E department

	12	Death Rate	Total number of deaths occurred in A& E department Exclude- Brought in dead cases	Total number of patients attended in A&E department	(Number of deaths occurred in A&E department*100/Total number of cases attended in A&E)	Monthly	Emergency register/Death register	Indicator for Quality of A&E department
Service Quality Indicator	13	LAMA Rate	Total number of patients leave against medical advice from A&E department Exclude- Abscond, Referral cases	Total number of patients attended in A&E department	(Number of LAMA patients*100/Total number of cases attended in A&E)	Monthly	Admission Register A& E Department	Indicator of service quality and patient satisfaction with treatment and stay in Hospital
	14	Absconding Rate	Total number of patients left the facility without any information provided to hospital staff Exclude- LAMA, Referral	Total number of patients attended in A&E department	(Number of absconding patients *100/Total number of cases attended in A&E Department)	Monthly	Admission Register A& E Department	Indicator of service quality and patient satisfaction with treatment and stay in Hospital
COP 9b	15	Incidence of patient falls	Number of patient falls	Total number of inpatient days	(Number of patient falls/Total number of inpatient days)*1000	Monthly	Admission Register A& E Department	

OUTCOME INDICATORS								
	2. Out Door Patient Department							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Proportion of Follow up patients	Total number of follow up OPD visits	Total OPD attendance Inclusion:- Old and New	Total old cases visited the OPD for follow up /Total patient attended in OPD for the month	Monthly	OPD Registration	Utilization of OPD by old patients
	2	ICTC OPD Per thousand (ICTC-Integrated Counselling and Testing Centre)	Total number of ICTC visits	Total Population= Total population in defined geographical area of the DH	(Total number of ICTC visits*1000/Total Population)	Monthly	ICTC Register	Utilization of ICTC services
	3	Proportion of BPL patients	Total number of BPL Patients Inclusion:- Patients having BPL card or equivalent card	Total OPD attendance Inclusion:- Old and New	(Total Number of BPL Patient registered in OPD/Total Number of patient Registered in OPD)	Monthly	OPD Registration	Indicator to see Utilization of OPD by BPL patients.
Efficiency	4	OPD Per Doctor	Total number of patients attended in each dept OPD	Number of Oncologists available in the hospital Inclusion: Regular, contractual, Part Time	Total number of Patient consulted in each OPD/Total number of doctor appointed for each department OPD	Monthly	OPD Register	Indicator for measuring efficiency of Medicine Doctor in OPD

	5	Consultation time at each OPD Clinic	Total Consultation time (Total number of medicine clinic days*number of minutes medicine clinics)	Total Medicine clinic visits	Total number of clinic days in a month's* Functional hours in a day*60/Total visits in OPD	Monthly	OPD Register	Indicator for measuring clinical care at General Medicine OPD
Service Quality Indicator	6	Waiting time at New Registration OPD	Waiting time at New Registration OPD will be the moment patient entered in Queue to finally being consulted by MO		Mean of waiting time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30)	Monthly	Time Motion Study	To observe waiting time at General OPD
	7	Average door to Drug time	Time taken by a patient from entering in queue for OPD registration to finally getting drugs at drug dispensing counter		Mean of waiting time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30)	Monthly	Time Motion Study	To observe overall OPD Delivery services
	8	Percentage of Oncology patients who had treatment initiated following multidisciplinary meeting. (Tumour Board)	Number of new Oncology patients who had treatment initiated following multidisciplinary meeting(tumour board)	Number of new oncology cases(all disciplines)treated in the month	[Number of new Oncology patients who had treatment initiated following multidisciplinary meeting(tumour board)/ Number of new oncology	Monthly		

					cases(all disciplines)treated in the month] * 100			
PSQ 2c	9	Waiting time for outpatient consultation	Sum total time (in minutes)for consultation	Total Number of outpatients	Sum total time (in minutes)for consultation/ Total Number of outpatients	Monthly		
COP.9.b	10	Incidence of patient falls	Number of patient falls	Total number of inpatient days	(Number of patient falls/Total number of inpatient days)*1000	Monthly		

OUTCOME INDICATORS								
	3. Paediatric Ward							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Bed occupancy rate	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Product of Total number of functional beds and days in the month	(Total Patient bed days in Paedia ward *100/Functional bed in paedia ward*days in month)	Monthly	Mid night census	Indicator for Utilization of Paediatric ward
	2	Proportion of mothers given nutritional counselling	Total number of mothers/attendant counselled for nutrition	Total number of admission in paedia ward	Total number of mothers or attendants counselled for nutrition/Total admission	Monthly	Counselling Register	Indicator for Utilization of counselling services
	3	Number of paediatric admission per 1000 indoor admission	Total admission in paediatric ward	Total admission of Hospital	(Total number of Paediatric cases admitted*1000/Total number of admission in the hospital)	Monthly	Admission Register	Indicator for Utilization of paediatric services
	4	Proportion of female patient	Total number of female child admitted in paediatric ward	Total admission in Paediatric ward	Total Number of female Patient admitted in paedia ward/Total admission in paedia ward	Monthly	Admission Register	Indicator to see Utilization of paediatric services by female patients

	5	LAMA Rate for female patient	Total female child leaves against medical advice from paediatric ward Exclude- Abscond, Referral, cases	Total admission of female child	(Total Number of female LAMA Patient in paedia ward *100/Total female admission in paedia ward)	Monthly	Admission Register & Discharge Register	Indicator to see Utilization of paediatric services by female patients
	6	Proportion of BPL Cases	Total number of BPL Patients admitted in paediatric ward Inclusion:- Patients having BPL card/equivalent card	Total admission in Paediatric ward	(Total Number of BPL Patient admitted/Total Number of patient admitted in Paedia ward)	Monthly	Admission Register	Indicator to see Utilization of paediatric services by BPL patients.
Efficiency	7	Referral Rate	Total number of patients referred to other centre from Paedia ward Exclude:- LAMA, Abscond	Total admission in Paediatric ward	Total number of Patient referred from paedia ward/Total number of Patient admitted in Paedia ward	Monthly	Transfer/Discharg e Register	Indicator for measuring efficiency of clinical process
	8	Bed Turn over rate	Total number of discharges Inclusion:-Normal discharge, LAMA, Referral, Abscond, Deaths	Total number of beds available in paedia ward	Total number of discharges from the paedia ward/Total number of beds	Monthly	Transfer/Discharg e Register	Indicator for measuring efficiency of clinical process

	9	Number of Drugs stock out in the paediatric ward			Total Stock outs days:- Stock outs events of essential drugs added for the month Inclusion – List of Commodities Exclusion – Strikeout of any other drug (One stock out day means one drug not available in the ward for one day)	Monthly	Stock Register	Indicator for measuring efficiently availability of essential drugs in the ward
	10	Discharge Rate	Total number of discharges Exclusion:-LAMA, Referral, Abscond, Deaths	Total admission	Total number of discharges in the paedia ward/Total number of admission in the Paedia ward	Monthly	Transfer/Discharge Register of Paedia Ward	Indicator for measuring efficiency of clinical process
Clinical care and safety	11	No. of new-born/child resuscitated			Total number of Resuscitated child for the month to be reported	Monthly	Case sheet	Indicator to measure Quality of resuscitation services
	12	Average length of stay	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Total no. of Discharges in the Months Inclusion-LAMA, Death, Referral and Absconding in the Month	All Patient bed days in paedia ward/Total number of discharges	Monthly	Mid night census & Transfer Register	Indicator of Quality of Clinical care and infection control practices

	13	Death Rate	Total number of deaths occurred in paedia ward Exclusion:- Brought in deaths	Total admission	(Total number of deaths in the paediatric ward*100/Total number of admission in the paediatric ward)	Monthly	Admission Register & Death Register	Indicator to measure Quality and safe paediatric health services
	14	Number of adverse events per 1000 patients	Total number of adverse events Inclusion:-Needle stick injury, child theft/swapping, Mother and child falls, Medication error, animal/rodent bites	Total admission	(Total number of incidences of adverse events *1000/ Total Number of admission)	Monthly	Adverse event form/Register & Admission register	Indicator to measure safe clinical practises in Paediatric ward
	15	Time taken for initial assessment	Time taken from the moment the patient entered in paedia ward to the moment the patient's assessment done by Medical officer		Mean of time taken for initial assessment. Method:- observed for Derived sample through time motion study done on sample basis (at least 5% patients but not less than 30)	Monthly	Time motion study	Indicator for measuring Quality of initial care treatment
	16	Case fatality rate	Case fatality Rate:-The case fatality rate (CFR) is a measure of the severity of a disease and is defined as the proportion of reported cases of a specified disease or condition which are fatal within a specified time		(Total number of deaths due to particular case*100/Total number of deaths)	Monthly	Case sheet	Indicator to measure Quality of care in paediatric services of any particular case

Service Quality Indicator	17	LAMA Rate	Total number of patients left against medical advice from paedia ward Exclude- Abscond, Referral, cases	Total admission	(No. of LAMA Patients from paedia ward*100/Total no. of admission in paedia ward)	Monthly	Discharge & Admission Register	Indicator of service quality and patient satisfaction with treatment and stay in Paediatric ward
	18	Attendant Satisfaction score	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each patients in Patient satisfaction survey for Paediatric ward done each month on statistically adequate sample (at least 30)	Monthly	Record of Patient Feedbacks	Indicator of attendant satisfaction in Paediatric ward
COP.9.b	19	Incidence of patient falls	Number of patient falls	Total number of inpatient days	(Number of patient falls/Total number of inpatient days)*1000	Monthly		
AAC 3a	20	Time for initial assessment of indoor patients	Sum of time taken for the assessment(in minutes)	Total number of admissions	(Sum of time taken for the assessment (in minutes)/Total number of admissions)	Monthly		

MOM 3a	21	Incidence of medication errors	Total number of medication errors	Number of Patient Days	(Total number of medication errors/ Number of Patient Days) *100	Monthly		
COP.9.b	22	Incidence of hospital-associated pressure ulcers after admission	Number of patients who develop new/worsening of pressure ulcer	Total number of inpatient days	(Number of patients who develop new/worsening of pressure ulcer/Total number of inpatient days)*1000	Monthly		
PSQ 2a	23	Catheter-associated Urinary tract infection rate	Catheter- associated UTIs in a month	Number of urinary catheter days in that month	catheter- associated UTIs in a month/Number of urinary catheter days in that month)*1000	Monthly		
IPC1f	24	Compliance to hand hygiene practices	Total number of actions performed	Total number of hand hygiene opportunities	(Total number of actions performed/Total number of hand hygiene opportunities)* 100	Monthly		

PSQ.2.c	25	Time taken for discharge	Sum of time taken for discharge (in minutes)	Number of patients discharged	Sum of time taken for discharge (in minutes)/ Number of patients discharged	Monthly		

OUTCOME INDICATORS								
	4. Laboratory							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Number of HIV test done per 1000 population	Total number of HIV tests done in ICTC and Laboratory Exclude:- Point of care HIV test done in labour Room, OT ,wards etc.	Total population in defined geographical area of the DH	(Total no. of HIV tests done *1000/Total Population)	Monthly	Lab Register	Indicator to measure Utilization of HIV test services of Laboratory
	2	Number of Blood smear examined done per 1000 population	Total number of blood smear tests done in the laboratory	Total Population= Total population in defined geographical area of the DH	(Total no. of Blood smear tests done in the laboratory*1000/Total Population)	Monthly	Lab Register	Indicator to measure Utilization of blood smear services of Laboratory
	3	Number of AFB examined done per 1000 population	Total number of AFB tests done in the laboratory	Total Population= Total population in defined geographical area of the DH	(Total no. of AFB examined tests done in the laboratory*1000/Total Population)	Monthly	Lab Register	Indicator to measure Utilization of AFB services of Laboratory
	4	Lab test done per patient to OPD	Total number of tests done for OPD patients	Total patients registered in OPD(both old and new)	(Total number of lab tests done for OPD patients/Total number of OPD patients registered)	Monthly	Lab Register	Indicator to measure Utilization of laboratory services by OPD patients

	5	Lab test done per patients IPD	Total number of tests done for IPD patients	Total patients admitted in the Hospital	(Total number of lab tests done for indoor patients/Total number of admission in the month)	Monthly	Lab Register	Indicator to measure Utilization of laboratory services by indoor patients
	6	Proportion of lab test done at night	Total number of lab tests done during night time from 8PM to 8AM	Total number of tests done in the laboratory	(Total number of tests done in the night/Total number of lab test done in the laboratory)	Monthly	Lab Register	Indicator to measure Utilization of laboratory services during night time
	7	Proportion of lab test done for BPL patients	Total number of tests done for BPL patients Inclusion:- Patients having BPL card/equivalent card	Total number of tests done in the laboratory	(Total number of tests done for BPL Patients/Total number of lab test done in the laboratory)	Monthly	Lab Register	Indicator to measure Utilization of laboratory services by BPL patients
Efficiency	8	Number of test not matched in validation	Total number of tests which not matched in validation	Total number of tests sent for cross validation	Number of total tests not matched in validation/Total number of tests sent for cross validation	Monthly		Indicator to measure clinical efficiency of laboratory
	9	Z Score for Biochemistry(or Equivalent)			Median of all Z score or equivalent for tests included in EQAS	Monthly		Indicator to measure clinical efficiency of biochemistry lab

	10	Z Score for Haematology (or Equivalent)			Median of all Z score or equivalent for tests included in EQAS	Monthly		Indicator to measure clinical efficiency of Haematology lab
	11	Down time critical equipment	Total number of days equipment are not functional during the month Inclusion:-Auto/ Semi Auto Analyser, Blood Gas analyser, Incubator, Microscope, ELISA reader, Cell counter etc.		Total number of break down days added for SNCU	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of critical equipment
	12	Turn around time for routine lab investigations			Time elapsed between receiving of sample and deriving of results in routine case	Monthly	Lab Register	Timely reporting of laboratory tests improve patient care efficiency, effectiveness, and satisfaction
	13	Turn around time for emergency lab investigations			Time elapsed between receiving of sample and deriving of results in emergency case	Monthly	Lab Register	Timely reporting of laboratory tests improve patient care efficiency, effectiveness, and satisfaction

Clinical care and safety	14	% of Critical values reported within one hour	Total number of lab test results which are critical from patient point of view that can impact clinical decision and are reported within one hour	Total number of critical values of lab results detected in laboratory	(Number of critical values of lab results reported within one hour*100/Total number of critical values of lab results detected)	Monthly	Lab reporting register	Critical values reporting is considered an important laboratory process because it can impact clinical decision making, patient safety, and operational efficiency.
	15	Number of adverse events per thousand patients	Total number of adverse events like needle stick injury, Patient falls, Medication error, animal/rodent bites, chemical splash, wrong cross matching, etc.	Total number of patients in laboratory during month	(Total number of adverse events detected in laboratory*1000/Total number of patients in lab)	Monthly	Adverse event reporting data	Indicator for measuring adverse events
	16	Test Demography	Total number of test done in haematology or biochemistry or microbiology or cytology or clinical pathology	Total number of tests done in the laboratory	Proportion of lab test done in Haematology or Biochemistry or Serology or Microbiology or cytology or clinical pathology	Monthly	Lab Register of different department of laboratory	Indicator for measuring demography of individual department of laboratory

	17	Report correlation rate	Total tests which clinically correlated with lab test results during audit and feedback from clinician	Total number of cases audited and feedback received	(Total number of lab tests values correlated with clinical findings*100/Total number of cases audited + feedback received)	Monthly	Audit and feedback record	Indicator to measure quality of lab reporting
	18	Proportion of false positive/false negative	Total number of false positive or false negative test result diagnosed from rapid diagnostic kit	Total number of tests done by rapid diagnostic kit	(Total number of false positive or false negative cases detected for rapid diagnostic kit/Total number of tests done by rapid diagnostic kit)	Monthly	Lab Register	Indicator to measure quality of rapid diagnostic kit
Service Quality Indicator	19	Waiting time at sample collection area	Waiting time is time taken for a patient from joining the queue for sample collection to the actual time of sample collection		Mean of turn around time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30	Monthly	Time motion study	Indicator for measuring service Quality during routine working time
	20	Number of stock out incidences of Reagents	Number of stock out days for all reagents available in laboratory during the month. (Stock out days means number of days particular reagent are not	Total types of reagents available in the laboratory	(No. of Stock out days for all reagents*100/ Total types of reagents *Days in Month)	Monthly	Stock Register	Indicator for measure availability of reagents

			available)					

OUTCOME INDICATORS								
	5. Operation Theatre							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Proportion of other emergency surgeries done in the night	Total emergency major surgeries conducted during night (8 PM to 8 AM) Exclude- C-Section	Total Major surgeries conducted (Day+Night)	Total number of emergency surgeries conducted in night/Total number of major surgeries conducted	Monthly	OT Register	Preparation of OT at night time for emergency cases

	2	Number of major surgeries done per 1 lakhs population	Total Major surgery conducted	Total Population= Total population in defined geographical area of the DH	(Total number of major surgeries conducted*100000/Total Population)	Monthly	OT Register	Indicator to measure utilization of OT services for major surgical procedure
	3	CSSD/TSSU Productivity index	Total number of packs(Drums, Trays, kits etc.) sterilised	Total number of autoclaves available in CSSD/TSSU	Number of Packs Sterilized/Total number of autoclaves available	Monthly	OT Register & CSSD/TSSU Register	To check utilization of CSSD/TSSU services
	4	Down time critical equipment	Total Number of days Critical equipment were not functional during the month. Critical equipment are those whose no functionality may hamper deliveries of services or compromise safety for example , Boyles apparatus, Cardiac monitor/Pulse Oximeter ,OT Light, Radiant warmer, Diathermy/Electric cautery, C-arm ,Oxygen concentrator, etc.	Total number of critical equipment * Total days in the month	Total number of days critical equipment were not functional during the month/Total number of critical equipment*Total days in the month	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of critical equipment

Efficiency	5	Skin to skin time	Sum of individual time taken from incision to suturing for all C-Section	Total number of C-Section conducted	Sum of time taken from incision to suturing for all C-section/Total number of C-section conducted	Monthly	Surgery note	Indicator for measuring efficient C-Section Operation
	6	Number of major surgeries per surgeon	Total number of major surgeries conducted	Total number of surgeon appointed. Inclusion:- all contractual, regular, on deputation, EmONC trained doctors	Total No. of Major Surgeries conducted/ Total No. of surgeon appointed in Hospital	Monthly	OT Register/HR Record	Indicator of efficiency of Surgeons and Operation theatre
	7	Proportion of emergency surgeries	Total number of emergency major Surgeries conducted in the OT	Total number of major surgeries conducted in OT	Total number of Emergency major surgeries conducted/Total number of major surgeries conducted	Monthly	OT Register	Indicator to measure efficient emergency surgical Procedure
	8	Cycle time for instrument processing	Time taken from cleaning to the completion of autoclaving process.		On sample basis	Monthly	CSSD/TSSU Register	Indicator to measure efficiency of instrument processing in OT

	9	Surgical site infection Rate	Total number of Surgical site infection detected (Any purulent discharge, absess, spreading cellulitis at surgical site during the month after the surgery)	Total number of surgeries conducted (major & minor surgeries)	(Total number of surgical site infection detected*100/Total number of surgeries Conducted)	Monthly	Infection control monitoring report	Indicator for Quality of infection control practices in OT
	10	Number of adverse events per thousand patients	Total number of adverse events like needle stick injury, deaths on OT table,, wrong drug administration, wrong site/wrong patient surgeries, retained instruments in patient discovered after surgical procedure, Patient falls, anaphylactic reactions etc.	Total number of surgeries conducted (major & minor surgeries)	Total number of incidences of adverse events*1000/ Total Number of surgeries conducted	Monthly	Adverse event form/Register & Admission register	Indicator to measure safe clinical practises in OT
Clinical care and safety	11	Incidence of re-exploration of surgery	Total number of operated cases returned to OT after complication	Total number of major and minor surgeries conducted	(Number of incidences of re-exploration after surgery*100/Total number of surgeries conducted)	Monthly	OT Surveillance register	Indicator to measure safe infection control practices in the OT
	12	% of environment swab culture reported positive	Total number of swab culture reported positive	Total number of swabs sent for culture	(Number of swab culture reported positive*100/Total number of swabs sent for culture from OT)	Monthly	OT Surveillance register	Indicator to measure safe infection control practices in the OT

	13	Perioperative death rate	Total number of deaths related to surgeries. Inclusion:- Pre operative, Operative and Post operative procedure before discharge	Total number of major & minor surgeries conducted	Total number of deaths related to surgeries/ Total number of surgeries conducted	Monthly	Death Register	Indicator to measure safe Surgical clinical protocols
	14	Proportion of General anaesthesia to spinal anaesthesia	Total number of surgeries conducted under General Anaesthesia	Total number of surgeries conducted under Spinal Anaesthesia	Total number of surgeries conducted under General anaesthesia/Total number of surgeries conducted under spinal anaesthesia	Monthly	OT Register	Indicator to measure Quality of anaesthesia services
	15	Proportion of PAC done out of total elective surgeries	Total number of elective surgeries where PAC done before shifting patient to OT	Total number of elective surgeries conducted	Total number of elective surgeries PAC done before shifting patient to OT/Total number of elective surgeries done	Monthly	OT Register	Indicator to measure quality of PAC services before surgery
	16	Number of autoclave cycle failed in bowie dick test out of total autoclave cycle	Total number of autoclave cycle failed in bowie dick test	Total number of autoclave cycle conducted	Total number of autoclave cycle failed in bowie dick test/Total number of autoclave cycle	Monthly	CSSD/TSSU Register	Indicator to measure Quality of sterilization services
Service Quality Indicator	17	Operation cancellation rates	Total number of operation cancelled after scheduling (Because of any reason)	Total number of surgeries conducted (Major & minor surgeries)	(Number of cancelled operation*100/Total surgeries conducted)	Monthly	OT Register	Indicator to measure service delivery of OT

COP 8d	18	Percentage of surgeries where the organisation's procedure to prevent adverse events like the wrong site, wrong patient, and wrong surgery have been adhered to.	Number of surgeries where the WHO safe surgery checklist was followed	Number of surgeries that were audited	(Number of surgeries where the WHO safe surgery checklist was followed/Number of surgeries that were audited)*100	Monthly		

OUTCOME INDICATORS								
	6.Intensive care unit							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Bed occupancy rate	ICU bed days- midnight head count of each day added for the month of all patients	Total number of functional beds available in ICU multiplied by number of days in the month	(Total Patient bed days in ICU*100/Functional bed in ICU*days in month)	Monthly	Mid night census	Indicator for Utilization of ICU
	2	Proportion of BPL patients admitted	Total number of BPL Patients admitted in ICU Inclusion:- Patients having BPL card/ equivalent card	Total admission in ICU	(Total Number of BPL Patient admitted in ICU/Total Number of patient admitted in ICU)	Monthly	ICU Admission Register	Indicator to see Utilization of ICU by BPL patients.
Efficiency	3	Down time critical equipment	Total number of days equipment are not functional during the month Inclusion:- Multi Para Monitor, Pulse Oximeter, ABG Machine, Ventilator, Defibrillator, Infusion Pump, C-PAP Bi-PAP, Suction Machine, Oxygen flow meter etc.	Total number of days in the month	Total number of break down days added for ICU equipment	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of critical equipment
	4	Referral Rate	Total number of patients referred from ICU Exclusion:- LAMA, Abscond	Total admission in ICU	Total number of Patient referred from ICU/Total number of Patient admitted in ICU	Monthly	Transfer/Referral Register of ICU	Indicator for measuring efficiency of clinical process

	5	Re admission Rate	Total number of patients readmitted. (Readmission:- Readmission to the ICU within 24 hrs of transfer during a single hospital stay)	Total admission in ICU	Total number of Patient readmitted /Total number of Patient admitted in ICU	Monthly	ICU Admission Register	Indicator of efficiency of Clinical services of ICU
Clinical care and safety	6	Average length of stay	Patient Census of each day added for the month	Total no. of Discharges in the Months Inclusion-LAMA, Death, Referral and Absconding in the Month	Total Patient bed days in ICU/Total number of discharges from ICU	Monthly	Mid night census & Transfer Register	Indicator of Quality of Clinical care and infection control practices during ICU stay
	7	Risk adjusted mortality rate/ Standard Mortality rate	The Standardized Mortality Ratio (SMR) is defined as the ratio of the observed mortality rate to the expected mortality rate. This permits performance-based comparisons of ICUs by adjusting for disease category and severity of physiological derangement. The reference values for the expected mortality rates are obtained by documenting mortality rates of patients from a large number of ICUs in a specific population. These are then stratified based on	All patients admitted in intensive care units	Risk-adjusted Mortality 1 = (Observed Rate/Risk-adjusted expected Rate) X100 Observed rate = Actual death in ICU/institution. Risk adjusted expected rate = Predicted death rate by predictive Model	Monthly	Hospital record for the observed mortality	Risk adjusted mortality rate/Standard mortality rate is a very useful parameter, often used to compare outcomes in two or more groups under study. It also gives an opportunity to individual ICUs for improving the processes

			disease categories and within multiple outcome score bands of standard ICU scoring systems. If the SMR for an ICU is <1, then the outcomes for that unit are interpreted to be better than the overall outcomes of the reference set used to develop the scoring system. Alternatively, an SMR of >1 signifies that the observed mortality rate is higher than the expected mortality rate, suggesting that the quality of care needs to be improve					and techniques.
	8	Number of pressure ulcer developed per thousand cases	Total number of Pressure ulcers developed in ICU	Total admission in ICU	Number of pressure ulcers*1000 / Total admission in ICU		ICU Records	Indicator to measure safe clinical practises in ICU
	9	Number of adverse events per thousand patients	Total number of adverse events like needle stick injury, child theft/swapping, falls, Medication error, animal/rodent bites	Total admission in ICU	(Total number of incidences of adverse events*1000/ Total Number of patient bed days in ICU)	Monthly	Adverse event form/Register & Admission register	Indicator to measure safe clinical practises in ICU

	10	UTI Rate	Total number of UTI cases detected in catheterized patients (Urinary tract infection (UTI) is an infection in any part of urinary system - kidneys, ureters, bladder and urethra.	Total number of catheter day(Sum total of catheter days of all catheterized patients)	(Total number of UTI detected*100/Number of catheter days	Monthly	ICU Data	Indicator to measure safe infection control and adherence to clinical protocol practices in the ICU
	11	VAP Rate	Total number of patients with VAP. VAP:- Ventilator associated Pneumonia **Radiologic Signs** ≥ 2 serial chest radiographs with at least one of the following: *New or progressive and persistent infiltrate *Consolidation *Cavitations **Clinical Signs** (at least one of the following): *Fever (temperature >38°C) with no other recognized cause *Leucopoenia (<4.0 × 109 cells/L) or leukocytosis (>12.0 × 109 cells/L) *For adults ≥ 70 y of age, altered mental status with no other recognized cause	Number of days mechanically ventilated with endotracheal tube	Number of patients with VAP*100/Number of days mechanically ventilated with endotracheal tube	Monthly	ICU Data	Indicator to measure safe infection control and adherence to clinical protocol practices in the ICU

	12	Adverse events are identified	Total number of adverse events like needle stick injury, Patients falls, Medication error, animal/rodent bites, Decubitus Ulcer etc.		Total number of adverse events	Monthly	ICU data	Indicator to measure safe infection control and adherence to clinical protocol practices in the ICU
	13	Reintubatio n rate	Total number of cases where reintubation done within 48 hours of extubation	Total number of extubation	(Number reintubated/ Number extubated)X 100	Monthly	ICU Data	Indicator to measure safe infection control and adherence to clinical protocol practices in the ICU
	14	Culture surveillance sterility rate	Total number of swab culture reported positive	Total number of swabs sent for culture during the month	(Number of swab culture reported positive*100/Total number of swabs sent for culture from ICU for the month)	Monthly	ICU Surveillance register	Indicator to measure safe infection control practices in the ICU
Service Quality Indicator	15	LAMA Rate	Total number of patients left against medical advice from ICU Exclude- Abscond, Referral cases	Total admission	(No. of LAMA Patients from ICU*100/Total no. of admission in ICU)	Monthly	Admission /Discharge Register	Indicator of service quality and patient satisfaction with treatment and stay in ICU

	16	Patient satisfaction score	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each patient/attendant in Patient satisfaction survey for ICU done each month on statistically adequate sample (at least 30)	Monthly	Record of client Feedbacks	Indicator of Patient satisfaction in ICU
COP.9.b	**17**	Incidence of Patient fall (Falls per 1000 patient days)	Number of patient falls	Total number of patient days	(Number of patient falls/ Total number of patient days) *1000	Monthly	ICU Data	Indicator to measure adherence to protocols practiced in the ICU
IPC1f	**18**	Compliance to hand hygiene practices	Number of actions performed	Total number of hand hygiene opportunities	Number of actions performed/Total number of hand hygiene opportunities)* 100	Monthly		
PSQ 2a	**19**	Catheter-associated Urinary tract infection rate	Number of urinary catheter- associated UTIs in a month	Number of urinary catheter days in that month	(Number of urinary catheter- associated UTIs in a month/Number of urinary catheter days in that month)*1000	Monthly		

MOM 3a	**20**	Incidence of medication errors	Total number of medication errors	Number of Patient Days	(Total number of medication errors/ Number of Patient Days) *100	Monthly		

OUTCOME INDICATORS								
	7. Indoor Patient Department							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Bed occupancy rate of Medical wards	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Product of Total number of functional beds in Medical ward and days in the month	(Total Patient bed days *100/Functional beds*days in month)	Monthly	Mid night census	Indicator for Utilization of medical ward
	2	Bed occupancy rate for Surgical wards	Total Patient bed days (midnight head count of each day added for the month of all patients)	Product of Total number of functional beds in surgical ward and days in the month	(Total Patient bed days *100/Functional beds*days in month)	Monthly	Mid night census	Indicator for Utilization of Surgical ward
Efficiency	3	Referral Rate	Total number of patients referred from IPD Exclusion:- LAMA, Abscond	Total admission in IPD	Total number of Patient referred from ward*100/Total number of Patient admitted in the ward	Monthly	Transfer Register of IPD	Indicator for measuring efficiency of clinical process
	4	Bed Turn over rate	Total number of discharges Inclusion:- LAMA, Referral, Abscond, Deaths	Total number of beds available in IPD	Total number of discharges from the IPD/Total number of beds	Monthly	Transfer Register of IPD	Indicator for measuring efficiency of clinical process

	5	Discharge Rate	Total number of discharges Inclusion:- LAMA, Referral, Abscond, Deaths	Total admission in IPD	Total number of discharges in the IPD/Total number of admission in the IPD	Monthly	Discharge & Admission Register of IPD	Indicator for measuring efficiency of clinical process
	6	Number of Drugs stock out in the ward	Stock outs occurred for essential commodities each day added for the month Inclusion – List of Commodities Exclusion – Strikeout of any other drug	Product of Total no. of Commodities and days in the month	(No. of Stock out days for Essential Commodities*100/ Total no. of commodities *Days in Month)	Monthly	Stock Register	Indicator for measuring efficiently availability of essential drugs in the ward
Clinical care and safety	7	Average length of stay for Medical ward	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Total number of discharges from medical wards Inclusion:- LAMA, Referral, Abscond, Deaths	Total Patient bed days of Medical ward/Total number of discharges from Medical ward	Monthly	Mid night census & Transfer Register	Indicator of Quality of Clinical care and infection control practices in medical ward
	8	Average length of stay for Surgical ward	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Total number of discharges from surgical wards Inclusion:- LAMA, Referral, Abscond, Deaths	Total Patient bed days of Surgical ward/Total number of discharges from Surgical ward	Monthly	Mid night census & Transfer Register	Indicator of Quality of Clinical care and infection control practices in Surgical ward

	9	Time taken for initial assessment	Time taken from the moment the patient entered in IPD to the moment the patient's assessment done by Medical officer/Nurse		Mean of response time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30)	Monthly	Time motion study	Indicator for measuring Quality of initial care treatment
Service Quality Indicator	10	LAMA Rate	Total number of patients left against medical advice from IPD Exclude- Abscond, Referral cases	Total admission	(No. of LAMA Patients from IPD*100/Total no. of admission in IPD)	Monthly	Admission /Discharge Register	Indicator of service quality and patient satisfaction with treatment and stay in IPD
	11	Patient Satisfaction score	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each patients in Patient satisfaction survey for maternity ward done each month on statistically adequate sample (at least 30)	Monthly	Record of Patient Feedbacks	Indicator of patient satisfaction in IPD
AAC 3a	12	Time for initial assessment of indoor patients	Sum of time taken for the assessment (in minutes)	Total number of admissions	(Sum of time taken for the assessment (in minutes)/Total number of admissions)	Monthly		

MOM 3a	**13**	Incidence of Medication errors (Medication errors per patient days)	Total number of medication errors	Number of Patients days	(Total number of medication errors / Number of Patients days) *100	Monthly	Incident register	Indicator to measure safe clinical practises in wards
COP.9.b	**14**	Incidence of hospital-associated pressure ulcers after admission	Number of patients who develop new/worsening of pressure ulcer	Total number of inpatient days	(Number of patients who develop new/worsening of pressure ulcer/Total number of inpatient days)*1000	Monthly		
PSQ 2a	**15**	Catheter associated Urinary tract infection rate	Number of urinary catheter associated UTIs in a month	Number of urinary catheter days in that month	(Number of urinary catheter associated UTIs in a month / Number of urinary catheter days in that month) *1000	Monthly	Ward Data	Indicator to measure safe infection control practices in the Wards
PSQ.2.a	**16**	Surgical site infection rate	Number of surgical site infections in a given month	Number of surgeries performed in that month	(Number of surgical site infections in a given month / Number of surgeries performed in that month) *100	Monthly	Ward Data	Indicator to measure safe infection control practices in the Wards

IPC1f	**17**	Compliance to Hand hygiene practice	Total no. of hand hygiene actions performed	Total no. of hand hygiene opportunities	(Total no. of hand hygiene actions performed / Total no. of hand hygiene opportunities) *100	Monthly	Ward Data	Indicator to measure safe infection control practices in the Wards
COP.9.b	**18**	Incidence of Patient fall (Falls per 1000 patient days)	Number of patient falls	Total number of patient days	(Number of patient falls/ Total number of patient days) *1000	Monthly	Incident Register	Indicator to measure adherence to protocols practiced in the Wards

OUTCOME INDICATORS								
	8. Blood Bank							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Number of blood unit issued per thousand population	Total number of blood units issued from blood bank	Total Population= Total population in defined geographical area of the DH	(Total number of blood units issued from blood bank*1000 /Total population)	Monthly	Blood Issue Register	Indicator for Utilization of blood bank

	2	% of units issued for the transfusion at facility	Total number of blood units issued for patients admitted in the hospital	Total number of blood units issued by the blood bank	(Total number of blood unit issued for patients admitted in the hospital)*100 /Total number of units issued by the blood bank	Monthly	Blood Issue Register	Indicator for Utilization of blood bank by the District Hospital
	3	Number of voluntary donation done per 1000 population	Total number of blood units received by blood bank through voluntary donation	Total Population= Total population in defined geographical area of the DH	(Total number of units issued through voluntary donation *1000/ Total Population	Monthly	Donor collection Record	Indicator to see volume of voluntary donation of blood in the blood bank
	4	Number of unit supplied to storage units			Total number of blood units supplied to linked blood storage units	Monthly	Blood Issue Register	Indicator to measure capacity of blood bank to supply blood units to blood storage centre
	5	Blood donation camps held			Total number of camps held during month by the blood bank	Monthly	Blood camp Register	Indicator to measure number of camps organised by blood bank

	6	Proportion of blood units issued in Emergency cases out of total units issued in the month	Total number of blood units issued in emergency cases for life saving measures	Total number of blood unit issued	Total number of blood issued in emergency cases/Total number of blood units issued	Monthly	Blood Issue Register	Indicator to measure capacity of blood bank to issue blood in emergency situation
	7	Number of blood units issued for free of cost			Total number of blood units issued from blood bank free of cost without any user charges	Monthly	Blood Issue Register	Indicator to measure utilization of blood bank facility by economical weak society or JSSK beneficiary
Efficiency	8	Down time critical equipment	Total number of days equipment are not functional during the month Inclusion:-Blood bags refrigerator , Deep freezer, Platelet agitators, Microscope, ELISA reader, RH Viewer, Analyser, Thawing machine, etc.		Total number of break down days added for blood bank equipment	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of critical equipment

	9	% of blood units discarded	Total number of blood units discarded during the month	Total number of blood units collected for the month	(Total number of units discarded in the month*100/Total number of blood units collected for the month)	Monthly	Record of blood discarded	Indicator for measuring efficiency of blood bank procedure
	10	% of units issued against replacement	Total number of blood units issued against replacement	Total number of blood units issued by the blood bank	(No. of Unit Issued against replacement*100 / Total number of blood units issued)	Monthly	Blood issue Register	Indicator for measuring availability of blood through voluntary donation
	11	% of unit tested seroreactive	Total number of blood units found sero reactive after testing Inclusion:- HIV, Hep A, Hep B, Malaria, VDRL	Total number of units collected	(Number of units found sero reactive*100/ Number of units collected)	Monthly	Record of processing of donor's blood	Indicator for measuring efficiently availability of essential drugs in the ward
Clinical care and safety	12	Blood transfusion reaction rate	Total number of transfusion reaction reported during the month Inclusion:- Minor and Major	Total number of blood units transfused	(Number of blood transfusion reactions*1000/Number of blood units transfused)	Monthly	Transfusion reaction report/form	Indicator of Quality of blood transfusion services
	13	Adverse events are identified and reported	Total number of adverse events identified Inclusion:-Chemical splash, needle stick injury, wrong cross matching, wrong blood group issue, Vaso vagal		All these types of events should be reported (Number only)	Monthly	Adverse event reporting data	Indicator for measuring adverse events

			shock etc.					
	14	Component to whole blood ratio	Total number of blood components issued	Total number of whole blood units issued	Number of blood components issued/Number of blood units issued	Monthly	Blood Issue Register	Indicator to measure Quality of blood component services
	14	Cross matched/Transfused ratio	Total number of blood units cross matched by request	Number of blood units actually transfused	Total number of units that are cross matched on request/Number of units actually transfused	Monthly	Cross matching Register	Indicator to measure safe blood issue practices
	15	% of single unit transfusion	Total number of incidents where in single unit issued	Total number of blood unit issued by the blood bank	(Number of single unit issued*100/Total number of units transfused)	Monthly	Blood Issue Register	Indicator to measure safety of blood transfusion
Service Quality Indicator	16	Time gap between issuing and requisition of blood in routine condition	Time Gap=Time taken from the moment request for blood is made to actual issue of blood units in routine condition	Sample of at least 5 patient per months	Average time taken from the moment patient request for blood to actual time of issuing of blood , observed in time motion study done at variable time on sample basis (at least 5 patients)	Monthly	Time motion study	Indicator for measuring service Quality during routine working time

	17	Time gap between issuing and requisition of blood in emergency condition	Time Gap=Time taken from the moment request for blood is made to actual issue of blood units in emergency condition	Sample of at least 5 patient per months	Average time taken from the moment patient request for blood to actual time of issuing of blood , observed in time motion study done at variable time on sample basis (at least 5 patients)	Monthly	Time motion study	Indicator for measuring service Quality during emergency situation
	18	Donor Satisfaction score of Blood Bank	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each donor in Patient satisfaction survey for blood bank done each month on statistically adequate sample (at least 30)	Monthly	Record of Donor Feedbacks	Indicator of Donor satisfaction in Blood Bank

OUTCOME INDICATORS - IMAGING								
	9. Radiology							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	X-ray done per 1000 OPD patients	Total number of x-ray done for OPD Patients	Total OPD attendance Inclusion:- Old and New	(Total no. of x-ray done for OPD patients*1000/Total OPD patient)	Monthly	Radiology Register	Indicator to measure Utilization of X-ray services by OPD Patients
	2	X-ray done per 1000 IPD patients	Total number of x-ray done for IPD Patients	Total Admission	(Total no. of x-ray done for indoor patients*1000/Total IPD patient)	Monthly	Radiology Register	Indicator to measure Utilization of X-ray services by IPD Patients
	3	Ultrasound done per 1000 OPD patients	Total number of USG done for OPD Patients	Total OPD attendance Inclusion:- Old and New	(Total no. of USG done for OPD Patients*1000/Total OPD Patients)	Monthly	Radiology Register	Indicator to measure Utilization of X-USG services by OPD Patients
	4	Proportion of X-ray done at night	Total number of x-ray done during night time i.e. between 8PM to 8AM	Total number of X-ray done(Day+Night)	(Total no. of X-ray done at night/Total X-ray done)	Monthly	Radiology Register	Indicator to measure Utilization of X-ray services at night

	6	Proportion of BPL patients screened	Total number of x-rays done for BPL Patients Inclusion:- Patients having BPL card or equivalent card	Total number of x-rays done	Total number of X-rays done for BPL patients /Total number of X-rays done	Monthly	Radiology Register	Indicator to measure equity
Efficiency	7	Down time critical equipment	Total number of days equipment are not functional during the month Inclusion:-X-ray Machine, USG machine and Dental X-ray machine, Portable X-ray, C-Arm etc.		Total number of break down days added for radiology equipment	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of critical equipment
	8	Turn around time for X-ray film development	Time taken from exposure to final development of films		Time elapsed between exposure to final development of films on derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30	Monthly	Radiology Register	Indicator to measure efficiency of x-ray technician for film development
	9	Proportion of waste of films	Total number of films wasted Inclusion:- Wastes of films due to Over exposure, Under Exposure, incorrect technique or incorrect development process	Total number of film used during the month	Total number of films wasted /Total number of films	Monthly	Stock Register	Indicator to measure clinical efficiency of radiology department

	10	Proportion of x-ray rejected/repeated	Total number of X-rays rejected or repeated on request by treating doctor	Total number of X-ray done	Total number of films rejected or repeated by treating doctors/Total number of x-ray done	Monthly	Radiology Register	Indicator for measuring efficiency of X-ray film development
	11	X-ray done per radiographer	Total number of x-ray done	Total number of x-ray technician or radiographer appointed	Total number of x-ray done/Total number of technician or radiographer appointed	Monthly	Radiology Register	Indicator to measure efficiency of radiographers
Clinical care and safety	12	Proportion of X-ray for which report is signed by radiologist	Total number of x-ray reports signed by radiologist	Total number of x-ray reports generated	Total number of X-ray reports signed by radiologist/Total number of X-ray reports generated	Monthly	Radiology Register	Indicator to measure quality of x-ray reporting
	13	Proportion of scans for which F-form is filled out of pregnant women scanned	Total number of form F reported to the competent authority	Total number of USG done for pregnant women	Total number of F-form reported to the competent authority/Total number of USG done for pregnant women	Monthly	USG Register and F-Form	Indicator to measure adherence to rule governed by government
	14	Examination demography	Comparative Proportion of different types of X-rays like Chest X-ray, Abdominal X-ray KUB X-ray or special x-ray	Total number of x-ray done	Comparative proportion of different types of X-rays/Total number of X-ray done	Monthly	Radiology Register	Indicator to measure demography details of different examination

	15	Report correlation rate	Total reports where findings of X-ray reports are correlated with clinical diagnosis	Total number of reports audited(At least 30 reports)	(Total number of x-ray reports are correlated with clinical diagnosis*100/Total number of cases audited)	Monthly	Audit Report	Indicator to measure quality of x-ray reporting
	16	Number of adverse events per 1000 patients	Total number of adverse events like patients falls, wrong report, over exposure of radiation	Total number of patients attended in radiology department	(Total number of adverse events*1000/Total number of patient attended in radiology department)	Monthly	Adverse event Register	Indicator to measure adverse events in radiology department
	17	Number of events of over limit of radiation exposure	Total number of cases where overexposure done to staff in radiology department Source:- BARC report of TLD badges		Total number of cases where overexposure done to staff in radiology department	Monthly	Radiology Register	Indicator to measure quality of radiation exposure and patient safety
Service Quality Indicator	18	Average Waiting time at radiology	Waiting time :-Time taken from the moment the patient joined the queue to the moment X-Ray taken		Mean of waiting time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30	Monthly	Time motion study	Indicator for measuring service Quality of radiology department
	19	Average Waiting time at USG	Waiting time :-Time taken from the moment the patient stand in the queue for USG examination to the moment actual USG done		Mean of waiting time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients	Monthly	Time motion study	Indicator for measuring service Quality of radiology department

					but not less than 30			
	20	Number of stock out incidences of X-ray films	Number of stock out days of X-ray films. (Stock out days means number of days X-ray films are not available)	Product of total number of types of X-ray films and days in month	(No. of Stock out days of X-Ray films*100/ Total no. of types of X-ray films *Days in Month)	Monthly	Stock Register	Indicator for measuring availability of x-ray films

OUTCOME INDICATORS								
	10. Pharmacy							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Percentage of drugs available against essential drug list for OPD	Total number of essential drugs available at drug dispensing counter	Total number of drugs in EDL for OPD	(Total number of essential drugs available at drug dispensing counter*100/Total number drugs in EDL for OPD)	Monthly	Stock Register of Drug dispensing counter	Indicator to measure availability of essential drugs for OPD patients
	2	Percentage of drugs available against essential drug list for IPD	Total number of essential drugs available in indoor department	Total number of drugs in EDL for IPD	(Total number of essential drugs available in IPD *100/Total number drugs in EDL)	Monthly	Stock Register IPD	Indicator to measure availability of essential drugs for IPD patients
	3	Expenditure on drugs procured by local purchase for BPL patients			Total amount of money in INR spent on local purchase of drugs for BPL patients not available in the hospital	Monthly	Cash book of Local Purchase(BPL)	Indicator to availability of drugs for BPL patients and utilization of local purchase
Efficiency	4	Number of stock out situations in vital category medicines			Total number of Stock outs of vital drugs added for the month	Monthly	Stock Register	Indicator for measure availability of vital category of drugs

	5	Turn around time for dispensing medicine at pharmacy			Time taken between the patient/ attendants joined the queue at drug dispensing counter till the time they left pharmacy after taking medicine (observed in time motion study done at peak hours on sample basis -at least 5% patients but not less than 30)	Monthly	Time motion study	Indicator to measure efficiency of pharmacist
	6	% of expired during the months	Total number of drugs expired during the month	Total number of drugs available in hospital	(Total number of drugs expired *100/ Total number of drugs available in hospital)	Monthly	Stock Register of expired medicine	Indicator to measure wastage of drugs due to expired medicine
Clinical care and safety	7	Proportion of prescription found prescribing non generic drugs	Total number of prescription found with brand name during prescription audit	Total number of prescriptions audited during the month (at least 5% of prescriptions but not less than 30)	Total number of prescription found with brand name during prescription audit/Total number of prescription audited	Monthly	Prescription audit	Indicator to measure adherence to generic drugs prescription
	8	Number of adverse drug reaction per thousand patients	Total number of adverse drug reaction events reported	Total number of patients admitted in the hospital	(Total number of adverse drug reaction events reported*1000/Tota l number of patients admitted	Monthly	ADR Report	Indicator to measure Safe drug administratio n practices

	9	Percentage of irrational use of drugs/over prescription	Total number of prescription with irrational use of drugs/ over prescription during prescription audit	Total number of prescriptions audited during the month(at least 5% of prescriptions but not less than 30)	(Total number of prescription with irrational use of drugs or over prescription during prescription audit*100/Total number of prescription audited)	Monthly	Prescription audit	Indicator of prescription practices and adherence to Standard treatment Guidelines
MOM.3.f	**10**	Percentage of safe and rational prescriptions	Total number of safe and rational prescriptions	Total number of prescriptions audited	(Total number of safe and rational prescriptions/ Total number of prescriptions audited)*100	Monthly		Indicator to measure Safe drug administratio n practices

OUTCOME INDICATORS								
	11. Auxiliary Services							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Number of cases for which medical audit done			Total number of cases audited for medical audit (at least 5% of prescriptions but not less than 30)	Monthly	Medical Audit Record	Indicator to measure how many cases selected for medical audit
	2	Number of cases for which death audit done			Total number of death audits done (All deaths are required to be audited)	Monthly	Death Audit Record	Indicator to see how many death audit has been done
	3	Linen Index	Total number of bed sheets washed during the month Inclusion:- Only bed sheet of indoor patients Exclusion:- Table cloths, OT dress, Patients dress, Doctors dress, bed sheet used by hospital staff washed in laundry.	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Total number of bed sheets washed during the month/Patient bed days	Monthly	Linen Register and Mid night Census	Indicator to measure utilization of Laundry department of the hospital
	4	Diet Index	Total number of diets provided during the month for all patients Inclusion:- Normal	Total Patient bed days (Midnight head count of each	Total Number of diets provided during the month/Patient bed days	Monthly	Diet Register and Mid Night	Indicator to measure utilization of Dietary

			and Special diet	day added for the month of all patients)			census	department of the hospital
	5	Cycle for laundry services			Average time taken between the dispatch of dirty linen and receiving of clean linen	Monthly	Linen Register	Indicator to measure efficiency of Laundry department
	6	Proportion of Special diets	Total number of special diets served Inclusion:- Diabetic diet, liquid diet, semisolid diet, renal diet, low salt/potassium diet, low fat diet etc.	Total number of diets served during the month	Total number of special diets served/Total number of diets served	Monthly	Diet Register	Indicator to measure provision of special diets to the patients
Clinical care and safety	7	Medical Audit score	Sum of scores obtained in medical audits	Total number of medical audit conducted	Sum of scores obtained in medical audits/Total number of medical audits conducted	Monthly	Medical Audit Record	Indicator to measure clinical care
	8	Death Audit score	Number of death audits with preventable cause of death	Total number of deaths audited (All deaths to be audited)	Number of deaths audits with preventable cause of death/ Total number of deaths audited	Monthly	Death audit Record	Indicator to measure clinical care
Service Quality Indicator	9	Waiting time for getting handicap certificate			Average time taken from the day of request to actual receipt of handicapped certificate	Monthly	Handicap certificate issuing Register	Indicator for measuring service Quality for issuing handicap certificates

	10	Waiting time for getting death certificate			Average time taken from the day of request to actual receipt of death certificate	Monthly	Death Certificate issuing Register	Indicator for measuring service Quality for issuing Death certificates
	11	Patient feedback on cleanliness of linen	Sum of scores obtained in IPD patient satisfaction survey on attribute related to cleanliness of linen	Total number of respondents in IPD survey	Sum of scores obtained in IPD patient satisfaction survey on attributes related to cleanliness of linen/Total number of respondents in IPD Survey	Monthly	Record of Patient Feedbacks	Indicator to measure patient satisfaction with respect to linen service of hospital
	12	Patient feedback on Quality of food	Sum of scores obtained in IPD patient satisfaction survey on attribute related to Quality of food	Total number of respondents in IPD survey	Sum of scores obtained in IPD patient satisfaction survey on attributes related to Quality of food/Total number of respondents in IPD Survey	Monthly	Record of Patient Feedbacks	Indicator to measure patient satisfaction with respect to dietary service of hospital
IMS.2.d	**13**	Percentage of medical records having incomplete and/or improper consent	Number of medical records having incomplete and/ or improper consent	Number of discharges and deaths	(Number of medical records having incomplete and/ or improper consent / Number of discharges and deaths) *100	Monthly	Medical Audit Record	Indicator to measure clinical care

OUTCOME INDICATORS								
	12. Mortuary							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Proportion of non-MLC cases	Total number of dead bodies received without MLC	Total number of dead bodies received	Total number of dead bodies received without MLC/Total number of dead bodies received	Monthly	Mortuary Register	Indicator to measure cases of Non MLC
	2	Occupancy rate of cold Storage for dead bodies	Total dead body days in cold storage (total dead body count of each day added for the month)	Product of functional cold storage cabinet in mortuary and days in the month	Total dead body days in cold storage/ Functional cold storage cabinet*Days in the month	Monthly	Mortuary Register	Indicator to measure utilization of cold storage facility of mortuary
Efficiency	3	Mean Storage time for dead body in cold storage	Total dead body days in cold storage (total dead body count of each day added for the month)	Total number of dead bodies sent out of mortuary for final disposal	Total dead body days in cold storage/ Total number of dead bodies sent out of mortuary for final disposal	Monthly	Mortuary Register	Indicator to measure efficiency of cold storage equipment
	4	Down time of cold storage equipment	Total number of days equipment are not functional during the month Inclusion:-Cold storage cabinet etc.		Total number of break down days added for mortuary equipment	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of cold storage equipment

OUTCOME INDICATORS								
	13. General Administration							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Bed Occupancy Rate	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Product of Total number of functional beds in the hospital and days in the month	(Total Patient bed days *100/Functional beds*days in month)	Monthly	Mid night census	Indicator for Utilization of Hospital Indoor services
	2	Number of total admissions per thousand population	Total number of admissions	Total Population= Total population in defined geographical area of the DH	(Total number of admission in the Hospital*1000/Total Population)	Monthly	Admission Register of the Hospital	Indicator for Utilization of Hospital Indoor services by community
	3	IPD per thousand Population	Total number of admissions	Total Population= Total population in defined geographical area of the DH	(Total number of admissions in the Hospital*1000/Total Population)	Monthly	IPD Register	Indicator to measure utilization of hospital IPD services by community
	4	OPD Consultation Per thousand Population	Total number of OPD consultation by doctor	Total Population= Total population in defined geographical area of the DH	(Total number of OPD consultations by doctor*1000/Total Population)	Monthly	OPD Register	Indicator to measure utilization of hospital OPD services by community

	5	Number of beds per 10 thousand population	Total number of functional beds in the hospital Exclusion:- Observation beds in emergency, Labour table, Staff beds	Total Population= Total population in defined geographical area of the DH	(Total number of Functional beds of the hospital*10000/Total Population)	Monthly	Hospital Record	Indicator to measure hospital capability to serve the community
	6	Nurse to bed ratio	Total number of IPD admission	Total number of nurses appointed in the facility	Total number of IPD admission/Total number of nurses appointed	Monthly	Administration Record	Indicator to measure nursing clinical care services of the Hospital
	7	No. of meeting held under RKS			Number of RKS meeting held during the month	Monthly	RKS Register	Indicator to measure involvement of community participation
	8	Proportion of BPL patient in hospital	Total number of BPL Patients attended in Hospital Inclusion:- Patients having BPL card/equivalent card Inclusion: Both OPD and IPD	Total number of patients attended during the month Inclusion:- Both OPD and IPD cases	Total number of BPL patient attended in the hospital/Total Patients attended in the Hospital	Monthly	OPD and IPD Register	Indicator to measure Utilization of Hospital services by BPL.
Efficiency	9	Overall Referral rate	Total number of patients referred from the facility Inclusion:- Emergency and indoor cases Exclusion:- Referral from OPD	Total admission in the facility	(No of cases referred out from the hospital*100/ Total no. of cases admitted)	Monthly	Referral Register	Indicator for efficiency of clinical process

	10	Overall discharge rate	Total number of cases discharged from the Hospital during the month Exclusion:- Abscond, LAMA & Death	Total admission in the facility	No of cases discharged from the hospital*100/ Total no. of admission)	Monthly	Discharge Register	Indicator of adherence to new-born death audit process
	11	Proportion of fund/grant utilized	Total amount of funds granted till the last month Inclusion:- Funds granted from state or national budget	Total amount of funds utilized at the end of the month	Total amount of funds received till the last month/Total amount of funds utilized at the end of the month	Monthly	Cash book	Indicator to measure efficiency of Hospital for utilizing funds provided
	12	Average length of stay	Total Patient bed days (Midnight head count of each day added for the month of all patients)	Total number of discharges. Inclusion:- Normal discharge, LAMA, Abscond, Referral, deaths	Total Patient bed days /Total Discharges	Monthly	Admission and Discharge Register	Indicator of Quality of Clinical care and infection control practices
Clinical care and safety	13	Crude mortality rate	Total number of deaths	Total number of admission	Total number of deaths/Total number of admission	Monthly	Death Register	Indicator to measure clinical care of the Hospital
	14	Hospital acquired infection rate	Total number of Hospital acquired infections detected in the facility Inclusion:- Surgical site infection, Device related infections , Respiratory and blood born infections	Total admission in the facility	(Total number of Hospital acquired infections*100/Total number of admission)	Monthly	Infection control Register	Indicator to measure clinical care

	15	Overall LAMA Rate	Total number of LAMA patients from the facility Exclusion:- Abscond and referral cases	Total admission in the facility	(No. of LAMA Patients from the facility*100/Total no. of admission)	Monthly	Admission /Discharge Register	Indicator of service quality and patient satisfaction with treatment and stay in IPD
Service Quality Indicator	16	Patient Satisfaction Score IPD	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each patients in Patient satisfaction survey for indoor patients done each month on statistically adequate sample (at least 30)	Monthly	Record of Patient Feedbacks	Indicator of patient satisfaction in IPD
	17	Staff Satisfaction Score	Sum of *average satisfaction score* of each respondent (Average satisfaction score = sum total of scores of attributes/number of total attributes)	Total number of respondents	Mean of scores given by each staff in staff satisfaction survey for all staff	Yearly	Record of Staff Satisfaction survey	Indicator to measure Staff satisfaction with the Hospital
	18	Turnover rate of Contractual staff	Total number of contractual staff leaves the job every month	Total number of contractual posts sanctioned in the hospital(or Average number of staff available on 1st and last days of the month)	(Total number of contractual staff leaves the job every month/Total number of contractual posts sanctioned	Monthly	HR Record	Indicator to measure Satisfaction of contractual staff

OUTCOME INDICATORS								
	14. Nuclear Medicine Department (NMD)							
Type	**S No**	**Quality Indicator**	**Numerator**	**Denominator**	**Formula**	**Frequency**	**source of data**	**Significance**
Productivity	1	Imaging Test done per 1000 OPD patients	Total number of Imaging Tests done for OPD Patients	Total OPD attendance Inclusion:- Old and New	(Total no. Of imaging test done for OPD patients*1000/Total OPD patient)	Monthly	NMD Register	Indicator to measure Utilization of NMD services by OPD Patients
	2	Imaging Test done per 1000 IPD patients	Total number of Imaging done for IPD Patients	Total Admission	(Total no. Of imaging done for indoor patients*1000/Total IPD patient)	Monthly	NMD Register	Indicator to measure Utilization of NMD services by IPD Patients
	3	Proportion of diagnostics imaging done at night	Total number of diagnostics imaging done during night time i.e. between 8PM to 8AM	Total number of diagnostics imaging done(Day+Night)	(Total no. of diagnostics imaging done at night/Total diagnostics imaging done)	Monthly	NMD Register	Indicator to measure Utilization of NMD services at night
	4	Proportion of BPL patients screened	Total number of diagnostics imaging done for BPL Patients Inclusion:- Patients having BPL card or equivalent card	Total number of diagnostics imaging done	Total number of diagnostics imaging done for BPL patients /Total number of diagnostics imaging done	Monthly	NMD Register	Indicator to measure equity

Efficiency	5	Down time critical equipment	Total number of days equipment are not functional during the month		Total number of break down days added for NMD equipment	Monthly	Equipment Maintenance Register	Indicator for measuring efficiency of critical equipment
	6	Proportion of PET CT rejected/repeated	Total number of PET CT rejected or repeated on request by treating doctor	Total number of PET CT done	Total number of PET CT rejected or repeated by treating doctors/Total number of PET CT done	Monthly	NMD Register	Indicator for measuring efficiency of PET CT services
	7	Diagnostic Imaging done per technician	Total number of Diagnostic Imaging done	Total number of Diagnostic Imaging technician appointed	Total number of Diagnostic Imaging done/Total number of technician appointed	Monthly	NMD Register	Indicator to measure efficiency of NMD Technicians
Clinical care and safety	8	Proportion of report is signed by Faculty	Total number of reports signed by Faculty	Total number of reports generated	Total number of reports signed by Faculty/Total number of reports generated	Monthly	NMD Register	Indicator to measure quality of reporting
	9	Number of adverse events per 1000 patients	Total number of adverse events like patients falls, wrong report, over exposure of radiation	Total number of patients attended in nuclear medicine department	(Total number of adverse events*1000/Total number of patient attended in radiology department)	Monthly	Adverse event Register	Indicator to measure adverse events in NMD
	10	Number of events of over limit of radiation exposure	Total number of cases where overexposure done to staff in radiology department Source:- BARC report of		Total number of cases where overexposure done to staff in radiology department	Monthly	NMD Register	Indicator to measure quality of radiation exposure

			TLD badges					and patient safety
Service Quality Indicator	11	Average Waiting time at NMD	Waiting time :-Time taken from the moment the patient reported at the counter to the moment procedure was started		Mean of waiting time for the derived sample observed in time motion study done at peak hours on sample basis (at least 5% patients but not less than 30	Monthly	Time motion study	Indicator for measuring service Quality of radiology department